AKSHITH SAI KONDAMADUGU

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate (B.E., 2026) with hands-on experience in Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI systems, and cloud-based model deployment on AWS and GCP. Proficient in building end-to-end AI/ML pipelines, LLM-powered multi-agent workflows, and RESTful API backends. Experienced with PyTorch, TensorFlow, LangChain, LangGraph, CrewAI, Hugging Face, FastAPI, Docker, Amazon SageMaker, and Google Cloud Platform. Holds AWS certifications in Machine Learning Engineering and AI Practitioner.

EDUCATION

Bachelor of Engineering in Computer Science Engineering *2022 – 2026*
Maturi Venkata Subba Rao Engineering College, Hyderabad, Telangana *Specialization: IoT, Cyber Security & Blockchain*

INTERNSHIP EXPERIENCE

AI/ML Virtual Intern *Apr 2025 – Jun 2025*
Google Cloud Platform — via MVSR Engineering College

- Designed, trained, and evaluated supervised learning models in TensorFlow on GCP across regression, classification, and deep learning use cases, achieving measurable improvements in model accuracy through systematic hyperparameter tuning.
- Built a computer vision object-recognition pipeline using Convolutional Neural Networks (CNN), image preprocessing, data augmentation, and transfer learning techniques to improve generalization on held-out test sets.
- Executed end-to-end ML workflows: dataset preparation, feature engineering, cross-validation, inference, and result interpretation in a reproducible engineering format.
- Documented experiment findings, model metrics, and architecture decisions aligned with production data science project standards.

AI/ML Virtual Intern *Apr 2024 – Jun 2024*
Amazon Web Services (AWS) — via MVSR Engineering College

- Built and deployed machine learning models using Amazon SageMaker with built-in algorithms and custom training scripts on structured datasets; served models as REST API endpoints.
- Integrated Amazon S3 for versioned data storage and used CloudWatch and SageMaker Clarify for performance monitoring, data drift detection, bias analysis, and model explainability.
- Executed the complete ML lifecycle on AWS: data ingestion, EDA, feature engineering, model training, evaluation, API deployment, and post-deployment monitoring.
- Applied MLOps best practices including model versioning, fairness checks, bias detection, and explainability auditing in a cloud-native environment.

PROJECTS

DynamicBI: AI-Powered Autonomous Business Intelligence Platform `dynamicbi.akshithsai.co.in`
Tech Stack: Python, FastAPI, LangGraph, React (Vite), LangChain, FAISS, Facebook Prophet, Scikit-learn, Pandas, NumPy

- Architected an 11-stage LangGraph multi-agent pipeline that autonomously handles ingestion, data cleaning, KPI computation, chart generation, anomaly detection, time-series forecasting, and LLM-driven business insight generation.
- Implemented specialized agents for CSV, Excel, SQL, and MongoDB ingestion; automated data sanitization; KPI calculations; and anomaly detection.
- Integrated a RAG semantic layer using FAISS and Google Generative AI embeddings to enable natural-language Q&A over live datasets.
- Built a React (Vite) chat-style frontend with async pipeline progress streaming and a unified dashboard for KPIs, charts, forecasts, anomaly explanations, and AI-generated business insights.

MultiModal RAG: Intelligent Multimodal Document Q&A Engine `rag.akshithsai.co.in`
Tech Stack: Python, FastAPI, LangChain, ChromaDB, React (Vite), NVIDIA NIM, OpenRouter, LLM Integration

- Built a full-stack multimodal RAG application enabling intelligent question answering across PDFs, DOCX, PPTX, CSV, images, web URLs, and YouTube transcripts through a unified chat interface.
- Engineered a modular ingestion pipeline with file-type-specific loaders and processors for text, tables, and images; generated semantic embeddings using NVIDIA's `llama-nemotron-embed-1b-v2` model and indexed vectors in ChromaDB for fast retrieval.
- Integrated NVIDIA Llama 4 Maverick vision model for image and slide understanding, enabling true multimodal comprehension beyond text-only RAG systems.
- Implemented configurable Top-K retrieval with source citations — every answer surfaces the exact document chunks with modality, page number, and similarity score — and deployed a zero-config React (Vite) frontend with runtime API key configuration.

Handwritten Digit Recognition: Deep Learning on MNIST
Tech Stack: Python, PyTorch, Flask, NumPy, Matplotlib

- Designed and trained a CNN from scratch in PyTorch on the MNIST dataset achieving high accuracy on 10-class digit classification.
- Applied batch normalization, dropout regularization, ReLU activations, and Adam optimizer to reduce overfitting and accelerate convergence.
- Evaluated model using confusion matrices, precision/recall metrics, and learning curves; iteratively tuned architecture depth and hyperparameters.
- Deployed the trained model as a Flask REST API for real-time inference, returning predicted labels and confidence scores; visualized convolutional activations to interpret learned features.

Decentralized Banking DApp: Blockchain Transaction Platform
Tech Stack: Solidity, Web3.js, Node.js, MetaMask, Ethereum

- Built a decentralized application (DApp) on the Ethereum blockchain enabling users to perform secure deposit and withdrawal transactions entirely on-chain via smart contracts.
- Authored and deployed Solidity smart contracts handling transaction logic, balance tracking, and access control with gas-optimized code.
- Integrated MetaMask wallet for user authentication and transaction signing, providing a seamless Web3 browser-based experience.
- Connected the frontend to the Ethereum network using Web3.js and Node.js, ensuring real-time interaction with deployed contracts and transparent on-chain data storage.

TECHNICAL SKILLS

Programming Languages	Python, Java, C/C++, JavaScript
AI / Machine Learning	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, CNN, RNN, LSTM, NLP, Transfer Learning, Supervised & Unsupervised Learning, Model Fine-Tuning
Generative AI & LLMs	LangChain, LangGraph, CrewAI, Prompt Engineering, RAG (Retrieval-Augmented Generation), FAISS, ChromaDB, Google Gemini, NVIDIA NIM, LLM Integration, Multi-Agent Systems
Agentic AI Frameworks	LangGraph Agents, CrewAI Multi-Agent Pipelines, Autonomous AI Workflows, Tool-Use Agents, Orchestration
Cloud Platforms	AWS, Google Cloud Platform, Microsoft Azure
MLOps & DevOps	Docker, Git, GitHub, CI/CD, Model Versioning, Experiment Tracking, Bias Detection
Data Science & Analytics	Pandas, NumPy, Matplotlib, Seaborn, Facebook Prophet, Anomaly Detection, Time-Series Forecasting, EDA
Web & API Development	FastAPI, Flask, React (Vite), Node.js, Streamlit, REST APIs, HTML, CSS
Blockchain & Web3	Solidity, Web3.js, Ethereum, MetaMask, Smart Contracts, DApp Development
Databases	SQL, MongoDB, ChromaDB, FAISS
Developer Tools	VS Code, Jupyter Notebook, Postman, Linux

CERTIFICATIONS

- **AWS Certified Machine Learning Engineer – Associate** — *Amazon Web Services*
- **AWS Certified AI Practitioner** — *Amazon Web Services*
- **Data Science and Machine Learning** — *HarvardX — edX*
- **Multi AI Agent Systems with CrewAI** — *DeepLearning.AI*